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RASD by Hamidreza Khodadadi and
Mohammad Rashidkhan

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1 Introduction

1.1 Purpose

Urban cyclists often find it difficult to identify **safe, comfortable, and efficient routes** in their city. Information about bike lanes, traffic intensity, road quality, and obstacles (such as potholes or damaged surfaces) is usually incomplete, outdated, or scattered across many sources. At the same time, cyclists continuously generate valuable information while riding, such as GPS traces, speed profiles, and movements of their mobile devices, which typically remain unused. As a consequence, planning a pleasant and safe bike trip often relies on personal experience and trial-and-error, which may discourage people from using bikes as a regular means of transportation.

The **Best Bike Paths (BBP)** project aims to address these issues by providing a software system that helps cyclists **collect, maintain, and exploit an inventory of bike paths**. The system supports riders in recording their trips, annotating the quality of the paths they use, and reusing this knowledge to choose suitable routes between an origin and a destination. The ultimate purpose of BBP is to **enable cyclists to plan and perform safer and more effective bike trips** by relying on up-to-date, community-contributed information about bike paths. The purpose of the project is articulated into the following goals:

- **[G1 – Trip Recording]**

  Allow registered users to record their bike trips through a mobile device, capturing GPS traces and timestamps that describe the travelled route.

- **[G2 – Trip Statistics and History]**

  Provide users with statistics for each recorded trip (e.g., distance, duration, average speed) and maintain a personal history of their cycling activities, optionally enriched with relevant meteorological information, when available, from external services.

- **[G3 – Manual Path Definition]**

  Enable registered users to manually define bike paths by selecting streets or segments and associating them with properties such as path status (e.g., *optimal*, *medium*, *sufficient*, *requires maintenance*) and known obstacles (e.g., potholes, hazardous segments).

- **[G4 – Automated Path Data Acquisition]**

  Support an automated mode in which the system, while the user is biking, collects GPS data and sensor readings (accelerometer and gyroscope) to reconstruct the followed path and detect potential issues affecting path quality.

- **[G5 – Confirmation of Inferred Information]**

  Provide mechanisms for users to review, confirm, or correct the information inferred in automated mode (e.g., detected potholes), ensuring that only validated data becomes publishable.

- **[G6 – Path Inventory and Publication]**

  Maintain a structured inventory of bike paths and their associated information, and allow their owners to mark data as publishable so that it becomes available to all users of the system.

- **[G7 – Path Search and Visualization]**

  Allow users, including unregistered users, to request bike paths between a chosen origin and destination and visualize one or more candidate routes on a map.

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- **[G8 – Path Scoring and Recommendation]**

  Compute a score for each candidate path based on its status, obstacles, and effectiveness in reaching the destination from the origin, and use this score to rank alternative paths so that cyclists can easily choose the most suitable one.

Together, these goals define the purpose of BBP: **to turn cyclists' trips and feedback into actionable knowledge that improves the safety and quality of future bike routes.**

1.2 Scope

In the context of the *Best Bike Paths (BBP)* system, the relevant phenomena include:

1.2.1 World Phenomena

- **[WP1]**: Cyclists ride their bikes along existing streets and bike paths in the city, performing trips of varying duration and difficulty.
- **[WP2]**: The physical road infrastructure (streets, bike lanes, intersections) has properties such as surface quality, presence of potholes, and traffic intensity that may change over time.
- **[WP3]**: Weather conditions (e.g., rain, wind, temperature) influence the safety and comfort of cycling on given paths.
- **[WP4]:** Cyclists visually identify the status of the bike path (e.g., optimal, medium, sufficient).
- **[WP5]**: Cyclists plan journeys between an origin and a destination according to their own preferences (e.g., safety, comfort, travel time).
- **[WP6]**: Cyclists gain experience about the quality of the paths they use and may be willing to share this knowledge with others.

1.2.2 Shared Phenomena

World-Controlled Shared Phenomena:

- **[SP1]**: Users register to BBP, authenticate themselves, and manage their personal accounts through the application.
- **[SP2]**: Registered users start and stop the recording of their bike trips using their mobile devices.
- **[SP3]**: Users enable or disable the automated acquisition mode, granting or revoking permission for BBP to use location and motion sensors while they ride.
- **[SP4]**: Registered users manually define bike paths by selecting streets or segments in the application and assigning them a status and known obstacles.
- **[SP5]**: Users review, confirm, or correct information about path issues (e.g., potholes) that BBP has inferred from sensor data before this information becomes publishable.
- **[SP6]**: Users specify an origin and a destination and submit route queries to BBP.
- **[SP7]**: Users inspect the paths displayed by BBP on the map, compare alternatives, and choose which path to follow for their trip.

Machine-Controlled Shared Phenomena:

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- **[SP8]**: BBP records trip data (GPS positions, timestamps) and stores it as part of the user's trip history.
- **[SP9]**: BBP computes and presents statistics about each recorded trip, such as distance, duration, average speed, and similar indicators, possibly enriched with weather information obtained from external services.
- **[SP10]**: BBP reconstructs the path followed during a trip from GPS data and analyzes motion sensor readings to detect potential obstacles or surface problems.
- **[SP11]:** The system prompts the user to confirm or reject potential obstacles detected during the ride.
- **[SP12]**: BBP updates the inventory of bike paths with manually inserted or confirmed information, including path status and obstacles, and marks data as publishable or private according to the owner's choices.
- **[SP13]**: BBP processes a user's origin–destination query, selects candidate bike paths from the inventory, computes a score for each path based on its quality and effectiveness, and ranks them accordingly.
- **[SP14]**: BBP visualizes the selected paths on a map, together with their scores and relevant details, so that users can evaluate them.
- **[SP15]**: BBP enforces access control and privacy rules when returning trip and path information, exposing only what each user is allowed to see.
- **[SP16]:** The system sends requests to external map/weather APIs to retrieve necessary environmental context.

1.3 Definitions, Acronyms, Abbreviations

This section introduces the terminology used throughout the document. Its aim is to avoid ambiguity and to ensure consistent usage of terms.

1.3.1 Definitions

- **Bike Path:** A route intended to be suitable for bicycles, either because a proper bike lane exists or because traffic conditions (e.g., low car density, compatible speed limits) make the road appropriate for cycling.
- **Path Segment:** A portion of a bike path corresponding, for example, to a single street or a continuous stretch of roadway between two intersections.
- **Trip:** A cycling activity performed by a registered user and recorded by BBP, including the start and end times of the activity and associated statistics such as total distance and average speed.
- **Trip Point (GPS Sample):** A single geolocated point acquired during a trip, consisting of a timestamp and coordinates (latitude and longitude). Trip points form the sequence used by BBP to reconstruct the followed path.
- **Trip Statistics:** Quantitative metrics computed by BBP for a trip, such as total distance, duration, elevation trends, and average speed.

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- **Meteorological Information / Weather Condition:** Weather-related data associated with a trip (e.g., temperature, wind speed, precipitation, general weather conditions) retrieved from an external service.
- **Obstacle:** A relevant issue affecting a bike path, such as a pothole, rough surface, or other road-surface irregularity. Obstacles may be manually reported by users or created automatically after the confirmation of detected issues.
- **Detected Issue:** A potential obstacle inferred automatically by BBP from device sensor data (e.g., accelerometer, gyroscope) acquired during a trip. Each detected issue includes a type, location, and confidence level and must be confirmed, corrected, or rejected by the user.
- **Path Status:** A qualitative evaluation of the overall condition of a bike path (e.g., optimal, medium, sufficient, requires maintenance).
- **Manual Mode:** A mode in which registered users explicitly insert information about bike paths, selecting the streets composing the path and setting their status and known obstacles.
- **Automated Mode:** A mode in which BBP acquires data automatically from the user's mobile device while they are biking. The system infers that the user is biking based on speed, reconstructs the followed path using GPS data, and detects possible obstacles using accelerometer and gyroscope measurements.
- **Confirmation Phase:** A phase following automated acquisition in which the user is asked to confirm, correct, or discard issues inferred by BBP before this information becomes available to the community.
- **Publishable Information:** Information about bike paths provided to BBP (either manually or automatically) that the owner chooses to make available to all users of the system.
- **Registered User:** A user who has an account on BBP and can record trips, insert information about bike paths, confirm automatically detected issues, and publish information.
- **Non-Registered User:** A user who accesses BBP without logging in and can only request the visualization of bike paths between an origin and a destination.
- **Origin:** The starting geographic point of a route specified by the user in a route search query.
- **Destination:** The target geographic point of a route specified by the user in a route search query.
- **Route Query:** A request made by a user to BBP specifying an origin and a destination for which the system must visualize one or more bike paths.
- **Candidate Path:** A bike path evaluated by BBP in response to a route query. For each candidate path, BBP computes a score and displays it as one of the possible routes.
- **Path Score:** A numerical or ordinal value computed by BBP for a bike path based on the path's status, presence of obstacles, and overall effectiveness in reaching the destination from the given origin.

1.3.2 Acronyms

- **BBP**: Best Bike Paths.
- **GPS**: Global Positioning System.
- **RASD**: Requirements Analysis and Specification Document.

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- **UI**: User Interface.
- **API**: Application Programming Interface.

The acronyms UI and API are used in later sections of this document to describe external interfaces and are therefore listed here for completeness.

1.3.3 Abbreviations

- **DB**: Database.
- **ID**: Identifier.
- **Avg.**: Average (e.g., Avg. speed).
- **Min.**: Minimum.
- **Max.**: Maximum.

1.4 Revision History

Version	Date	Author(s)	Description
0.1	2025-11-22	Mohammad Rashidkhan	Prepare Introduction draft and add Purpose and Scope sections.
0.2	2025-11-23	Hamidreza Khodadadi	Add Definitions, Acronyms, Abbreviations, Reference and Structure sections.
0.3	2025-11-28	Mohammad Rashidkhan	Prepare Overall Description draft and add Product Perspective section.
0.4	2025-11-30	Mohammad Rashidkhan	Add Product Functions, Domain Assumptions, Dependencies, and Constraints.

Table 1: Revision history of the RASD.

1.5 Reference Documents

This document is based on the following material:

- The official RASD assignment specification for the Software Engineering 2 course, academic year 2025–2026, containing the problem definition “Best Bike Paths”.
- Slides and notes from the Software Engineering 2 course concerning requirements engineering and the preparation of RASD documents.

1.6 Document Structure

- **Section 1 – Introduction**: presents the purpose and scope of BBP, defines key terminology (definitions, acronyms, abbreviations), and reports revision history, reference documents, and the structure of this RASD.
- **Section 2 – Overall Description**: provides a high-level view of the BBP system, including the product perspective, main functions, user characteristics, assumptions, dependencies, and constraints.

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- **Section 3 – Specific Requirements**: details the functional and non-functional requirements of BBP, including external interface requirements, use cases and related diagrams, performance requirements, design constraints, and software system attributes.
- **Section 4 – Formal Analysis Using Alloy**: presents the formal model of selected aspects of BBP and the results of analyses performed using the Alloy tool.
- **Section 5 – Effort Spent**: summarizes the amount of work carried out by each group member for the preparation of this document.
- **Section 6 – References**: lists all external sources referenced in this RASD.

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2 Overall Description

This section provides a high-level overview of the Best Bike Paths (BBP) system. It outlines the context in which BBP operates and summarizes its overall perspective, key functions, intended users, and the main assumptions, dependencies, and constraints that shape the solution.

2.1 Product Perspective

2.1.1 Scenarios

A Cyclist Records a Trip Luca, a regular cyclist who often rides across the city, opens the BBP mobile application before starting his commute. After logging in, he taps the *Start Trip* button. BBP activates Luca's GPS and begins collecting his location data every few seconds. As he rides, the application also monitors the readings from his phone's accelerometer and gyroscope, registering significant vertical or lateral movements that may indicate bumps or potholes on the road.

During the trip, Luca checks the live dashboard, which shows elapsed time, current speed, and total distance covered. At some point, he rides through a rough segment of road; the phone vibrates noticeably, and BBP logs this sensor peak as a potential obstacle.

After reaching his destination, Luca taps *Stop Trip*. The system stores the trip in his personal history and enriches it with weather information (temperature, wind, rain intensity) retrieved from an external weather service. Later, Luca reviews detailed trip statistics such as distance, duration, and average speed. The final GPS position recorded when he stopped the trip is considered the destination of that trip.

A Cyclist Defines a Path Manually Maria wants to contribute to the community by documenting a bike-friendly route she frequently uses to reach her university. She logs into BBP and navigates to the *Manual Path Submission* section.

Here, she is presented with an interactive map. Maria selects the streets she uses, adding each segment to the path. For each selected segment, she assigns a path status such as:

- "Optimal" for recently renewed bike lanes,
- "Medium" for regular asphalt,
- "Sufficient" for a street where traffic is a bit heavier.

Along the route, Maria also marks relevant obstacles: a pothole near a particular intersection, a slippery segment on a bridge when wet, and a construction zone that temporarily narrows one of the lanes. After reviewing her inputs, she marks the path information as *publishable*, making it available to the rest of the BBP community.

A Cyclist Uses Automated Mode to Gather Path Information Alberto enables *Automated Mode* in the BBP app because he enjoys cycling through different parts of the city and wants BBP to automatically gather useful data from his rides. Once activated, the system continuously collects GPS positions, speed, and device motion signals, inferring Alberto's followed path.

During one of his rides, the accelerometer captures several sharp vertical movements when Alberto rides on cobblestones. BBP interprets these readings as possible road surface issues. However, the system knows such detections may include false positives; for example, the cyclist might simply drop the phone or hit a curb.

After the ride, BBP presents Alberto with a list of detected obstacles along with their map locations. For each detection, Alberto can:

- confirm it as a real pothole or bump,

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- correct the classification (e.g., “not a pothole, just cobblestones”), or
- reject the detection entirely.

Only after confirmation does BBP add the validated information to the path inventory. Alberto appreciates this workflow because it prevents incorrect data from being published.

A User Searches for Bike Paths Between an Origin and a Destination Elena, who is planning to ride from her home to a park across town, opens the BBP app without logging in. She goes to the *Find Path* feature and enters her origin and destination. BBP queries its inventory of publishable paths contributed by the community.

The system identifies several potential paths connecting the two points. Each path has an associated score, computed using:

- its overall status,
- the number and severity of obstacles along it,
- and its effectiveness in reaching the destination (for example, length and directness).

BBP then presents Elena with a set of routes on the map. The highest-scoring route is highlighted, while alternatives are also shown. Each route displays a summary of:

- expected distance and time,
- number of obstacles,
- whether segments are optimal, medium, or require maintenance.

Elena selects the highest-scoring route and starts navigating it using the map view provided by the app.

A Cyclist Reviews and Publishes Automatically Detected Information Sara rides daily to work using Automated Mode. After several days, BBP identifies recurring bumps at nearly the same location along her path. The system groups these observations and prompts Sara to validate them.

She opens the *Review Detected Issues* screen, which lists, for each candidate issue:

- location of the issue,
- frequency of detection,
- suggested issue category (e.g., pothole, uneven pavement).

Sara inspects the area on the map and confirms that the issue is indeed a pothole, which has been getting worse over the weeks. She marks the detection as correct and chooses to make the information publishable. The corresponding path segment’s status is updated to “requires maintenance”, improving BBP’s accuracy for other users.

2.1.2 Domain-level Class Diagram

The domain-level class diagram models the key entities and relationships that characterize the Best Bike Paths (BBP) application domain. It represents central concepts such as **Cyclist**, **Trip**, **Path**, **PathSegment**, **Obstacle**, **DetectedIssue**, and **RouteQuery**, along with their relevant attributes and the interactions among them.

The model captures how trips are recorded and reconstructed from GPS samples, how users manually or automatically contribute information about bike paths, how obstacles emerge and become publishable

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through user confirmation, and how BBP retrieves and ranks candidate paths for route queries. By focusing on world and shared phenomena rather than implementation details, the diagram provides a conceptual overview of how information flows within the domain.

A **Trip** represents a cycling activity recorded by a registered user. Each trip consists of a sequence of **TripPoints**, which are individual GPS samples collected during the ride. After the recording ends, BBP reconstructs the travelled route by geographically matching these TripPoints to the existing **PathSegments** in the inventory. This relationship is represented by the association indicating that a Trip may *traverse* several PathSegments.

During a trip, BBP may infer anomalies from accelerometer and gyroscope data, creating **DetectedIssues**. Each DetectedIssue is associated with exactly one Trip and is localized on exactly one PathSegment, reflecting where the potential obstacle occurred. After the ride, users must review the detected issues: confirmed or corrected ones result in the creation of new **Obstacles** on that segment, while rejected ones are discarded. Obstacles and paths can each be marked as *publishable*, determining the visibility of the information within the shared path inventory.

A **Path** is a route defined by a sequence of PathSegments, either manually specified by the user or derived from previously collected information. Each Path has a qualitative **status** selected from a finite set of values (e.g., optimal, medium, sufficient, requires maintenance). A Path includes at least one PathSegment, while each PathSegment may belong to one or more Paths.

Any user—registered or not—can request suggested bike routes by submitting a **RouteQuery**, specifying an origin and destination, each modeled as a **Location** value object. BBP evaluates candidate paths against the query and constructs **CandidatePath** entities, each referencing a Path and associated with a real-valued score computed from the path's condition, obstacles, and effectiveness in reaching the destination. Non-registered users are not explicitly represented in the diagram; their behavior is captured by RouteQuery instances without an associated Cyclist.

The following diagram provides a conceptual overview of the problem domain and avoids any implementation-specific detail.

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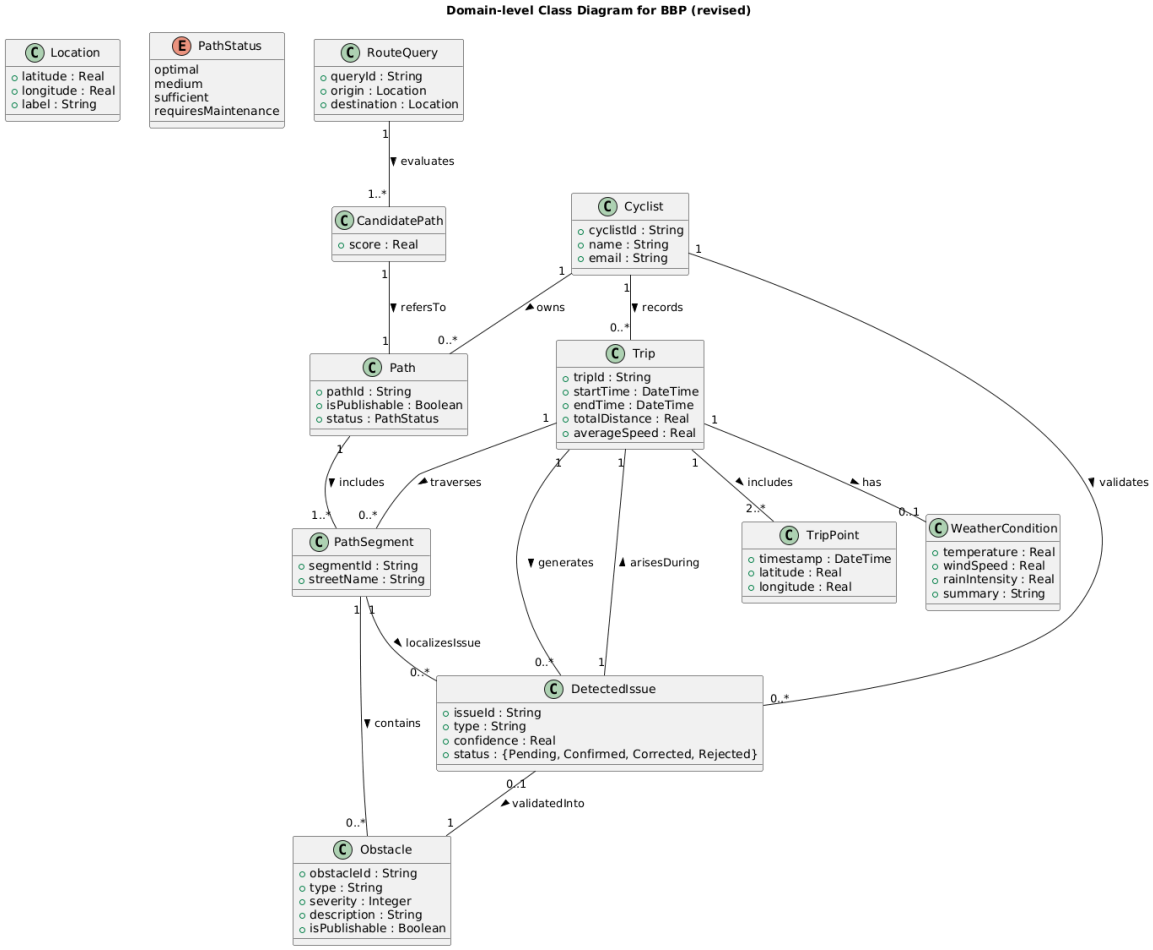


Figure 1: Domain class diagram

2.1.3 State Diagrams

State diagrams illustrate the dynamic behavior of key domain entities in BBP. They describe how Trips evolve during recording, how automatically DetectedIssues move through confirmation stages, and how Paths become publishable.

Trip State Diagram
This diagram represents the lifecycle of a recorded Trip. A trip begins when a cyclist starts recording. While the trip is ongoing, BBP collects GPS samples and sensor readings. When the cyclist stops recording, the system reconstructs the travelled route, computes statistics, optionally retrieves weather data, and stores the completed trip in the user's history.

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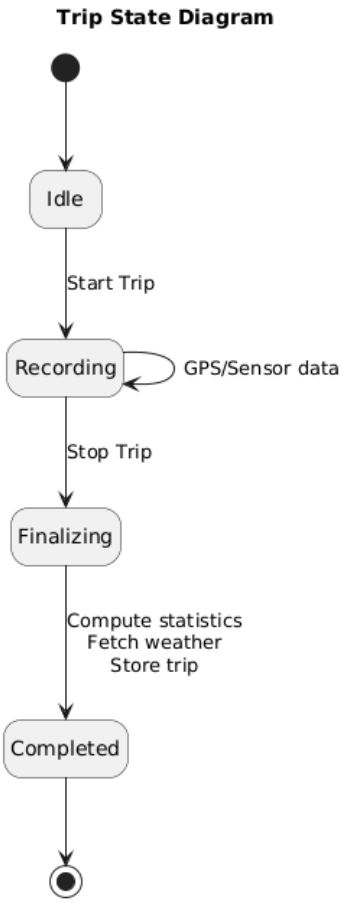


Figure 2: Trip State Diagram

Detected Issue State Diagram

This diagram models the lifecycle of a Detected Issue, that is, a potential obstacle inferred automatically from accelerometer and gyroscope anomalies during a trip. Since the system may produce false positives, each detected issue must go through a confirmation phase where the user validates, corrects, or rejects the inferred information. Confirmed or corrected issues become Obstacles; rejected ones are discarded.

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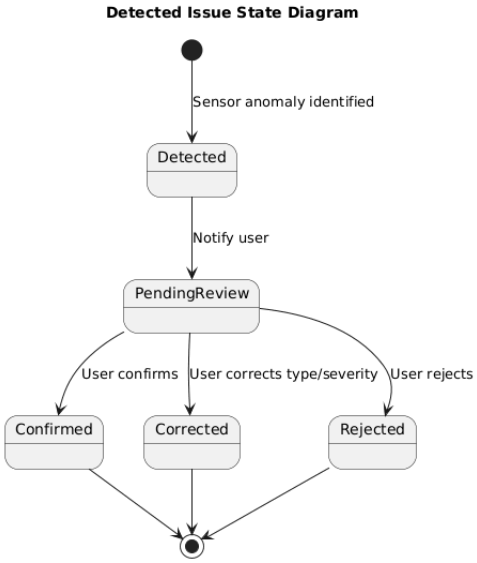


Figure 3: Detected Issue state Diagram

Path State Diagram

This diagram represents the lifecycle of a Path defined by a registered user. A path starts as a draft in manual mode, while the user selects segments and sets status. The path can then be saved as non-publishable (private), updated later, or made publishable so that all users can access it. Users may also update path information over time, leading the path back to the Editable state.

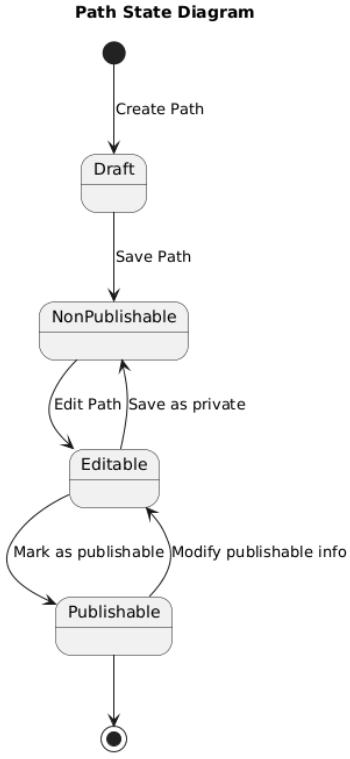


Figure 4: Path State Diagram

Obstacle State Diagram

This diagram represents the lifecycle of an Obstacle, i.e., a confirmed issue associated with a path seg-

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ment. An obstacle can be created either manually by the user or automatically after the confirmation of a detected issue. When created, the obstacle is initially private to its owner. The owner can then decide whether to publish or unpublish it, controlling its visibility in the shared path inventory.

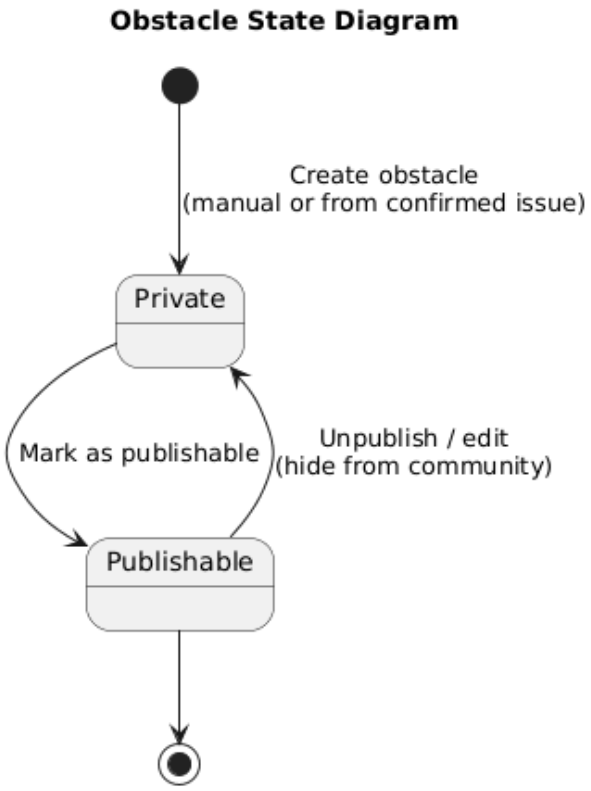


Figure 5: Obstacle State Diagram
Obstacle lifecycle from creation to publication.

RouteQuery State Diagram
This diagram represents the lifecycle of a **RouteQuery**, that is, a user request to visualize bike paths between an origin and a destination. A query is first composed by the user (setting origin and destination), then submitted to BBP for processing. The system retrieves candidate paths, computes their scores, and finally presents the ordered results on the map. After the answers are shown, the query reaches a completed state. Non-registered users issue queries in exactly the same way; they are not modeled explicitly here and are represented simply by RouteQuery instances without an associated Cyclist.

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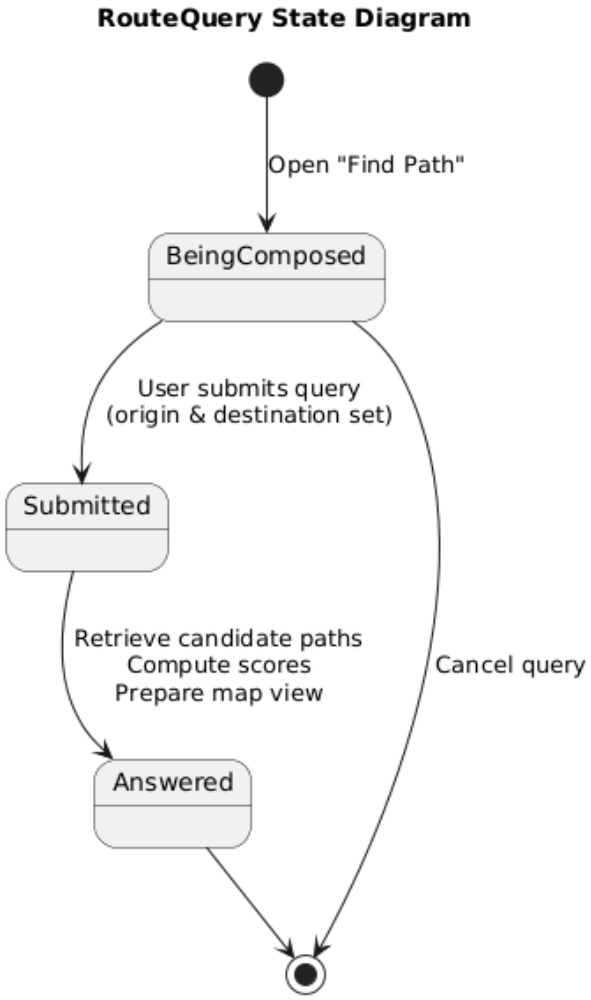


Figure 6: Route query lifecycle from composition to answer.

2.2 Product Functions

This section summarizes the main functions that BBP must provide in order to achieve the project goals defined in Section 1.1 and to support the scenarios in Section 2.1.

2.2.1 Trip Management

- **PF-T1 – Trip Recording**
  BBP shall allow a registered user to start and stop the recording of a bike trip from the mobile application. While recording, the system collects GPS samples and motion sensor data (accelerometer and gyroscope) at suitable intervals.

- **PF-T2 – Trip Reconstruction and Storage**
  After a trip is stopped, BBP shall reconstruct the travelled route from the collected GPS samples and store the trip in the user's personal history, together with its start/end times.

- **PF-T3 – Trip Statistics**
  For each recorded trip, BBP shall compute and store basic statistics such as total distance, duration, and average speed. These statistics shall be made available to the user in a trip-details view.

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- **PF-T4 – Weather Enrichment (When Available)**
  When an external weather service is reachable, BBP shall retrieve relevant meteorological information for the time and approximate area of the trip and associate it with the recorded trip.

- **PF-T5 – Trip History Browsing**
  BBP shall allow a registered user to browse their past trips, view trip details (including route and statistics), and select individual trips for inspection.

2.2.2 Manual Path Information Management

- **PF-P1 – Manual Path Definition**
  BBP shall allow a registered user to define a bike path manually by selecting streets or segments on a map and composing them into a Path.

- **PF-P2 – Path Status Management**
  For each user-defined path, BBP shall allow the owner to set and update the overall path status (e.g., optimal, medium, sufficient, requires maintenance).

- **PF-P3 – Manual Obstacle Annotation**
  BBP shall allow a registered user to associate obstacles (e.g., potholes, rough surface, dangerous crossings) to specific path segments, including information such as type, severity, and a short description.

- **PF-P4 – Publishable Path Information**
  BBP shall allow the owner of a path to mark the path information as publishable, so that its geometry, status, and publishable obstacles become part of the shared inventory of bike paths accessible to all users.

2.2.3 Automated Path Data Acquisition and Confirmation

- **PF-A1 – Automated Mode Activation**
  BBP shall allow a registered user to enable or disable an automated acquisition mode in which the system collects GPS and motion sensor data while the user is biking.

- **PF-A2 – Automatic Detection of Potential Issues**
  While automated mode is active and a trip is being recorded, BBP shall detect potential issues (DetectedIssues) based on sensor anomalies and associate them with the corresponding trip and path segment.

- **PF-A3 – Issue Review and Confirmation Phase**
  After a trip, BBP shall present the detected issues to the user and allow them to confirm, correct, or reject each issue.

- **PF-A4 – Conversion to Obstacles**
  For issues that are confirmed (possibly after correction), BBP shall create or update corresponding obstacles on the relevant path segments and, if the user chooses, mark them as publishable information

2.2.4 Path Search, Visualization and Recommendation

- **PF-R1 – Route Query Submission**
  BBP shall allow any user, whether registered or not, to specify an origin and a destination and submit a route query.

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- **PF-R2 – Candidate Path Retrieval**
  In response to a route query, BBP shall retrieve one or more candidate paths between the origin and the destination from the inventory of publishable paths and segments.

- **PF-R3 – Path Scoring and Ranking**
  For each candidate path, BBP shall compute a score that depends on the path status, the presence and severity of publishable obstacles, and the effectiveness of the path in reaching the destination from the origin. BBP shall order the candidate paths according to this score.

- **PF-R4 – Map Visualization of Paths**
  BBP shall display the candidate paths on a map, together with basic summary information (e.g., path length and qualitative score), so that users can visually compare and select a preferred path.

- **PF-R5 – Path Details Visualization**
  BBP shall allow the user to inspect a selected candidate path in detail, including the list and position of publishable obstacles along the path and the distribution of path statuses on its segments.

2.2.5 User and Access Management

- **PF-U1 – Registration and Authentication**
  BBP shall allow new users to register and existing users to authenticate, so that trips and owned path information can be associated with the correct Cyclist.

- **PF-U2 – Ownership and Visibility Control**
  BBP shall associate each path and piece of path information with its owner and enforce visibility rules so that only publishable information is visible to all users, while non-publishable information can be seen and modified only by its owner.

- **PF-U3 – Access for Non-Registered Users**
  BBP shall allow non-registered users to submit route queries and view publishable paths and obstacles but shall not allow them to record trips or create/modify path information.

2.3 User Characteristics

The BBP platform supports two primary types of end users: **registered cyclists** and **non-registered users**. In addition, **system administrators** manage technical and moderation tasks. Each user type has specific roles and characteristics when interacting with the platform.

2.3.1 Registered Cyclist

A registered cyclist is an individual who creates an account on BBP to record trips and contribute information about bike paths. Registered cyclists can start and stop trip recording from the mobile application, allowing the system to collect GPS and sensor data (accelerometer and gyroscope) during their rides. They can review their trip history, inspect trip statistics, and, when available, see associated meteorological information.

Registered cyclists are also responsible for enriching the path inventory. In **manual mode**, they can define bike paths by selecting streets or segments on the map, assign path statuses, and annotate obstacles on specific segments. In **automated mode**, they allow BBP to detect potential issues from sensor data and later participate in the confirmation phase by confirming, correcting, or rejecting detected issues. Furthermore, registered cyclists can decide which paths and obstacles become publishable so that other users can benefit from the shared information.

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2.3.2 Non-Registered User

A non-registered user is any person who accesses BBP without creating or logging into an account. These users are primarily consumers of the information collected by the community. They can open the application or web interface, specify an origin and destination, and request bike paths between these points. BBP presents them with one or more candidate routes, visualized on a map and ranked according to their quality and effectiveness.

Non-registered users cannot record trips, define paths, annotate obstacles, or participate in the confirmation of automatically detected issues. Their interaction is limited to browsing publishable paths and using the platform as a decision-support tool for planning bike routes.

2.3.3 System Administrator

A system administrator is responsible for the technical operation and maintenance of BBP. Administrators do not normally interact with the platform as cyclists but instead manage system-level aspects such as user account administration, monitoring of service availability, and configuration of external services (e.g., map and weather providers).

When necessary, administrators may moderate or remove path and obstacle information that is clearly incorrect, abusive, or violates legal or community guidelines. They also oversee data retention policies and ensure that stored trip and path data is handled according to privacy and security requirements.

2.4 Domain Assumptions, Dependencies, and Constraints

2.4.1 Domain Assumptions

- **[DA1: Internet and Device Access]**
  All users (registered and non-registered) have access to the internet and to a GPS-enabled mobile device (e.g., smartphone) with accelerometer and gyroscope sensors, capable of running the BBP application.

- **[DA2: Accurate Location and Sensor Readings]**
  The GPS, accelerometer, and gyroscope readings provided by users' devices are sufficiently accurate and frequent to reconstruct trips and to detect potential road-surface issues (e.g., potholes) with acceptable precision.

- **[DA3: Honest Path Information and Feedback]**
  Users are expected to provide honest and reasonably accurate information when manually defining paths, annotating obstacles, and confirming or rejecting automatically detected issues.

- **[DA4: Stable Road Network]**
  The underlying road network (streets and intersections) does not change dramatically over short periods, so that paths and path segments remain meaningful for a reasonably long time.

- **[DA5: Reliable Map Data]**
  The external map provider used by BBP supplies correct and up-to-date geographic information (e.g., street geometries and names) that can be relied upon when defining paths and matching trips to path segments.

- **[DA6: External Weather Service Availability]**
  When BBP requests meteorological information for a trip, the external weather service is assumed to be reachable and to return accurate weather data for the requested time and area.

- **[DA7: Reasonable Trip Duration and Battery Level]**
  Users start trip recording with a sufficient battery level and do not perform trips that are unusually

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long relative to their device's battery capacity, so that recording can run continuously without unexpected interruptions due to power loss.

- **[DA8: Single Rider per Device During Trip]**
  Each running instance of the BBP app is assumed to correspond to a single cyclist during a trip; devices are not shared among multiple cyclists while a trip is being recorded.

- **[DA9: Non-Malicious Intent]**
  The platform assumes that users do not deliberately generate fake trips, obstacles, or confirmations for the purpose of degrading the quality of the shared path information.

2.4.2 Dependencies

- **Map and Geocoding Services**
  BBP depends on external map and geocoding services to display maps, obtain street geometries, and associate GPS coordinates with streets and intersections when defining paths or reconstructing trips.

- **Weather Information Service**
  The system relies on an external weather provider to retrieve meteorological information (e.g., temperature, wind, precipitation) associated with recorded trips.

- **Cloud Hosting and Storage**
  BBP depends on a cloud-based infrastructure for hosting its backend services, storing user accounts, trips, paths, obstacles, and associated metadata, and for providing sufficient availability and scalability.

- **User Devices**
  The platform depends on compatible mobile devices (smartphones) equipped with GPS, accelerometer, and gyroscope sensors, and with a supported operating system version, for smooth interaction and data collection.

- **Notification / Messaging Services (if used)**
  If BBP sends push notifications or e-mails (e.g., reminders to review detected issues), it depends on third-party notification or e-mail services to deliver these messages to users.

2.4.3 Constraints

- **Supported Platforms**
  The initial release of BBP is intended to run on modern smartphone platforms (e.g., recent versions of Android and/or iOS). Other platforms (desktop browsers, legacy devices) are not initially supported.

- **Sensor and Sampling Limits**
  To preserve battery life and network usage, BBP must limit the frequency of GPS and sensor sampling and may batch or compress recorded data before transmission to the backend.

- **Data Storage and Retention**
  The system must store trip and path data in a way that respects reasonable storage limits. Very old or low-value raw sensor data may be aggregated or archived according to policies defined at design time.

- **Privacy and Data Protection**
  The collection and processing of location and sensor data must comply with applicable privacy

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regulations. Personal data (including trip histories) must be stored securely, and access to it must be restricted based on user identity and ownership.

- **Language Support**
  The initial version of BBP will support only one primary language (e.g., English). Additional languages may be considered as future enhancements.

- **Development Timeline**
  The system must be designed and implemented within the schedule defined for the course project, which constrains the amount of functionality and level of detail that can be implemented in the first release.

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3 Specific Requirements

3.1 External Interface Requirements

This section describes how the Best Bike Paths (BBP) system interacts with users, devices, other software systems, and communication channels.

3.1.1 User Interfaces

The BBP system provides different user interfaces tailored to the main categories of users: registered cyclists, non-registered users, and system administrators.

Cyclist Interface (mobile application). The mobile application is the primary interface for registered users while riding and when managing their personal cycling data. The main views are:

- **Login / Registration View**: allows a user to create an account or log in using email and password. It also provides access to password reset and to the option of using the application as a guest (limited features).
- **Home / Dashboard View**: displays a summary of the user's recent activity (e.g., last trips, total distance travelled), shortcuts to start a new recording, to open the trip history, and to define or edit paths.
- **Trip Recording View**: shows a map centred on the current position, current speed, elapsed time, and distance. It provides controls to start, pause/resume, and stop the recording. A status indicator reports whether automated mode (sensor-based obstacle detection) is active.
- **Trip History and Details View**: lists previously recorded trips with basic data (date, distance, duration). Selecting a trip opens a detailed page with the reconstructed path on the map, statistics (distance, duration, average speed), and, when available, meteorological information associated with the trip.
- **Manual Path Definition View**: provides an interactive map where the cyclist selects streets or segments to compose a bike path. For each segment the user can set the path status (e.g., optimal, medium, sufficient, requires maintenance) and annotate obstacles (potholes, rough surfaces, construction zones, etc.). The view allows the user to decide whether the information is private or publishable.
- **Detected Issues Review View**: lists issues detected automatically from accelerometer and gyroscope data during trips in automated mode. For each issue the user can see its location and type and can confirm it, correct its attributes, or reject it. Only confirmed issues can become part of the shared inventory.
- **Path Search and Visualization View**: allows both registered and non-registered users to specify an origin and a destination either by selecting them on the map or by entering textual addresses. The system displays one or more candidate paths on the map, each with a summary (distance, estimated travel time, path score, number and severity of obstacles). Users can compare these paths and choose one to follow.
- **Settings View**: lets users configure privacy options (default visibility of their contributions), manage permissions for location and motion sensors, and request deletion of their account and data.

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Guest Interface (mobile and web). Non-registered users can access a reduced set of features without authentication:

- access to the *Path Search and Visualization* view to look for bike paths between an origin and a destination;
- read-only access to path summaries and statistics;
- a prompt informing them that recording trips, defining paths, and confirming issues require registration.

Administrator Interface (web). System administrators access BBP through a web-based interface. Main functions include:

- monitoring system status and resource usage;
- viewing aggregated statistics about trips, paths, and obstacles;
- managing user accounts (e.g., disabling accounts in case of misuse);
- configuring connections to external services such as map and weather providers.

3.1.2 Hardware Interfaces

BBP interacts with the following hardware components:

- **User Mobile Devices**: smartphones or tablets used by cyclists must provide:
  - a GPS receiver (or equivalent location hardware) to acquire the sequence of positions during a trip;
  - an accelerometer and a gyroscope to detect significant movements used for identifying potential obstacles;
  - sufficient storage to hold temporary trip data when no network connection is available.

  The mobile application accesses these sensors via the standard APIs of the operating system and requires explicit user permission.
- **User Desktop / Laptop Computers**: used mainly to access the web interface. They must support a modern web browser and typical input devices (keyboard, mouse, or touchpad). No additional hardware is required.
- **Backend Servers**: one or more servers host the BBP backend components and databases. The RASD does not prescribe specific hardware, but the infrastructure must provide persistent storage and sufficient processing power to handle the expected user load.

3.1.3 Software Interfaces

BBP depends on several software interfaces:

- **Mobile Operating Systems**: the mobile application runs on major smartphone platforms (e.g., Android, iOS) and uses their APIs for:
  - location services (GPS and, when available, network-based positioning);
  - access to motion sensors (accelerometer, gyroscope);

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  - notification delivery;
  - handling of background execution and power-saving modes.

- **Web Browsers**: the web interface is implemented as a responsive web application and must operate correctly on recent versions of main browsers that support HTML5, CSS3, and modern JavaScript.
- **Backend REST API**: both mobile and web clients communicate with the server through a RESTful HTTP API. The API exposes operations for user authentication, trip upload, retrieval of trip histories, creation and modification of paths, search of paths between an origin and a destination, and management of detected issues. Requests and responses are encoded in a structured format (e.g., JSON).
- **Database Management System**: the backend uses a DBMS to persist users, trips, paths, obstacles, and system configuration data. All access to the DBMS is mediated by the backend application; clients never interact with the database directly.
- **External Map Service**: BBP integrates with a third-party map service through its API to display maps, convert addresses to coordinates, and obtain a base street network on which bike paths are defined. The service is expected to expose HTTPS endpoints and return data in machine-readable formats.
- **External Weather Service**: to enrich trips with meteorological information, the backend calls an external weather API using trip time and location. The service returns information such as temperature, precipitation, and wind speed.

3.1.4 Communication Interfaces

The communication channels used by BBP must satisfy the following requirements:

- **Network Protocols**: all communication between clients, backend, and external services uses HTTP or HTTPS. HTTPS is mandatory whenever authentication credentials or personal data are transmitted.
- **Data Formats**: the system uses standard formats such as JSON for data exchanged via the REST API. Map tiles and similar resources follow the formats defined by the selected map provider.
- **Notifications**: when necessary (e.g., to confirm that a trip has been saved or to remind the user to review detected issues), the backend uses platform-specific push-notification services or other mechanisms provided by the mobile operating system.
- **Offline Operation and Synchronization**: if network connectivity is not available during a trip, the mobile application stores recorded data locally. Once the connection is restored, the data are sent to the backend through the REST API. The server is responsible for resolving potential conflicts (for example, if a path is edited from two devices).
- **Security**: all communication channels carrying personal information or detailed trip data must use TLS. Expired or invalid certificates must be rejected, and authentication tokens must be protected during transmission.

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